

Advanced Animal Movement Modelling Training Workshop

# Machine Learning and Emerging Directions

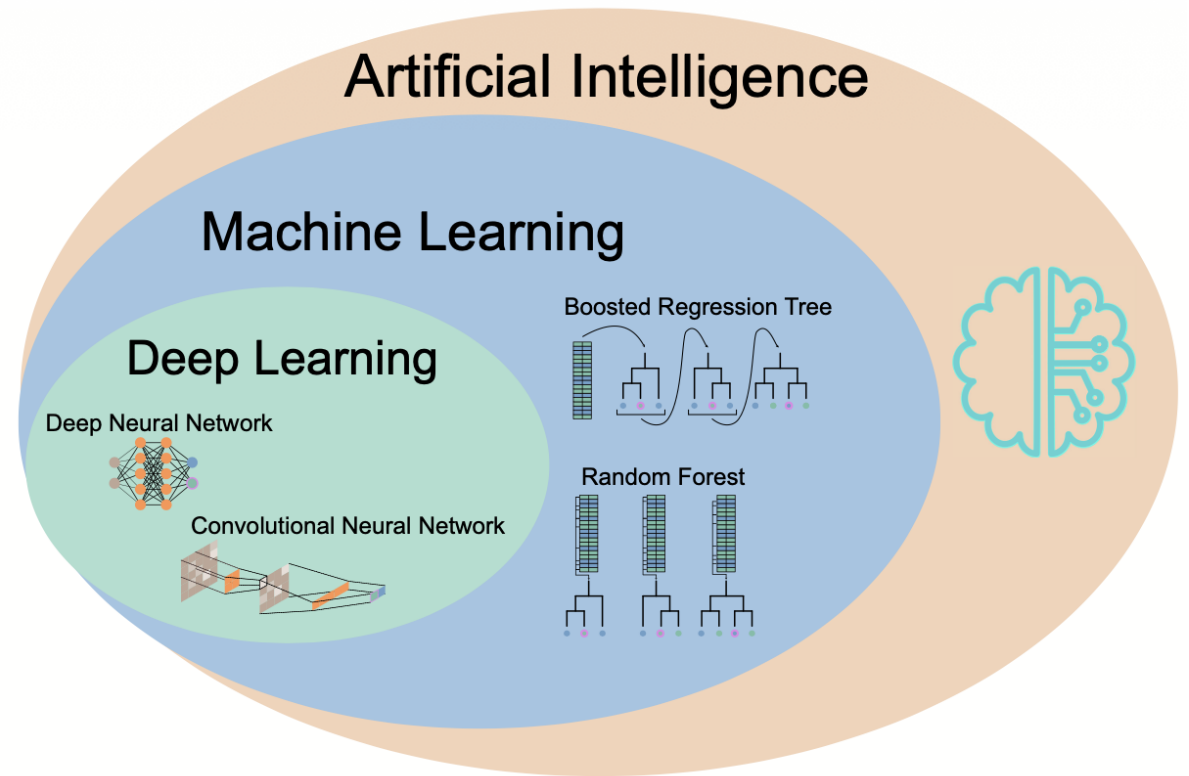
17 September 2026

Josh Cullen

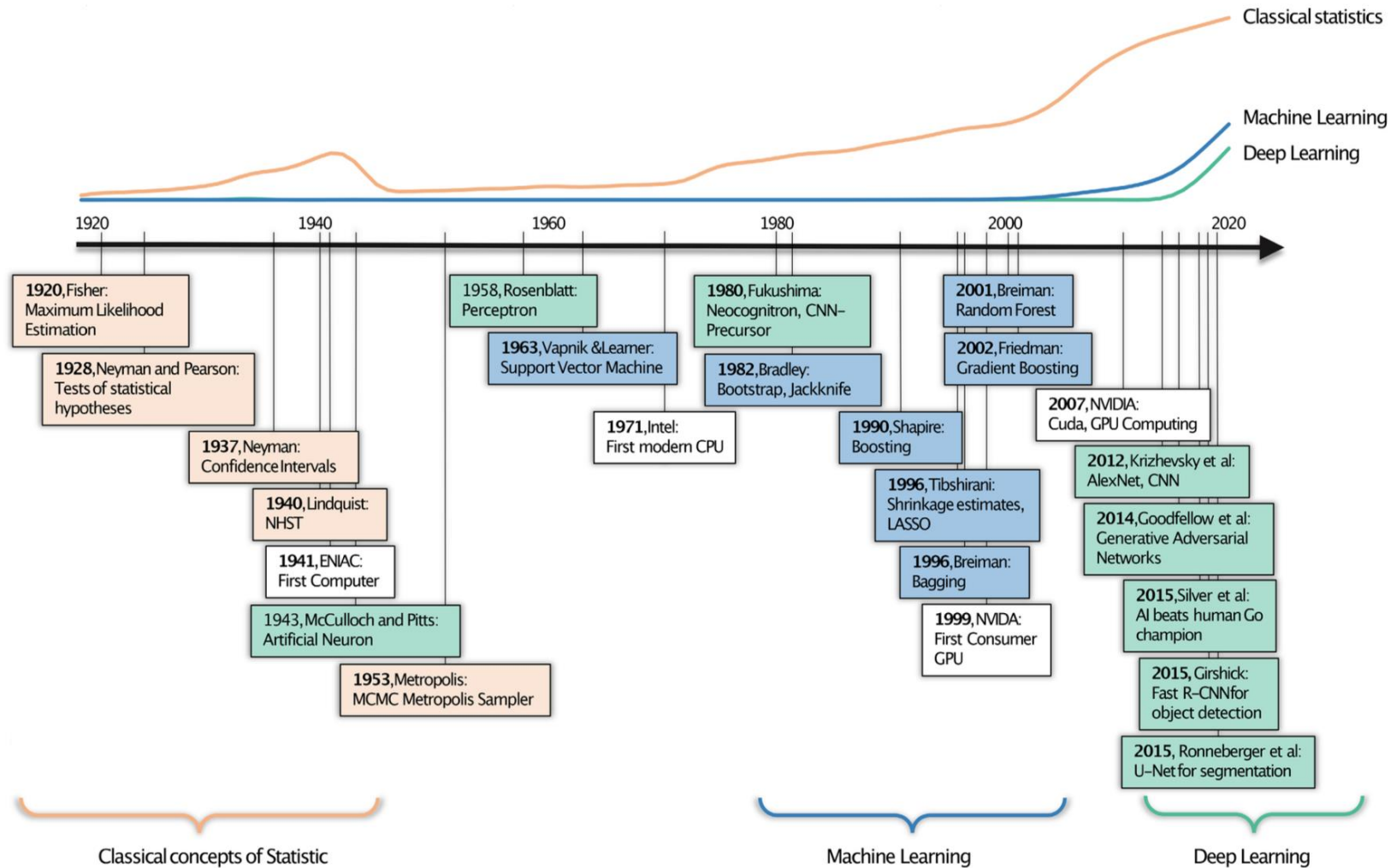


# What is Machine Learning?

- A family of algorithms that are trained for certain tasks by minimizing a loss function
- Considered distinct from traditional statistics in that they often don't include a data-generating model
  - i.e., no probability density functions or explicit assumptions
- Aims to use a selected algorithm to optimize bias-variance tradeoff

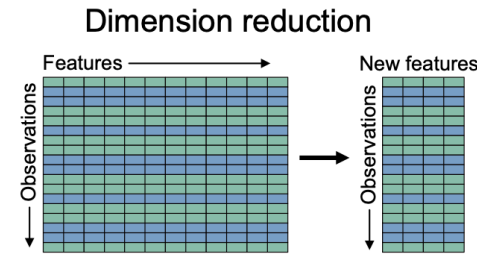


# Timeline of recent statistics



# General types of ML algorithms

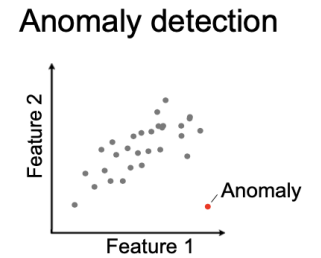
- Main dichotomy of **unsupervised** and **supervised** methods
  - Unsupervised methods don't have a response variable
  - Supervised methods include both classification and regression depending on whether response is discrete or continuous



- Principal component analysis (PCA)
- t-distributed stochastic neighbor embedding (tSNE)
- (Variational) Autoencoder

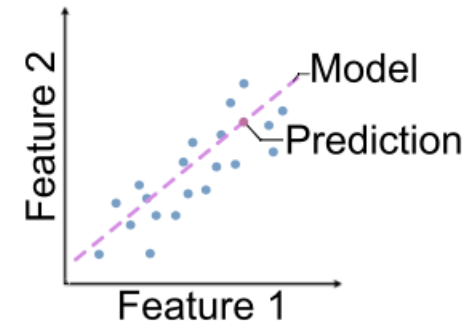


- *k*-means
- density based clustering (e.g. DBSCAN)
- hierarchical clustering
- Unsupervised random forest

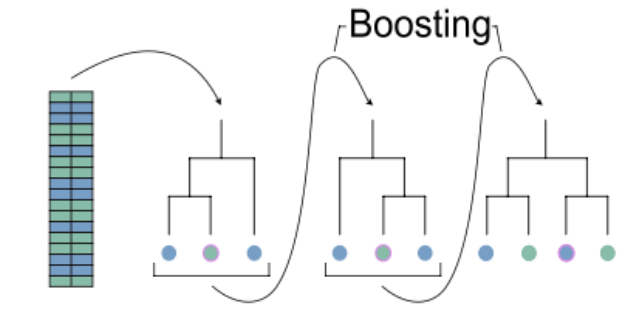


- (Variational) Autoencoder
- Clustering algorithms (e.g. *k*-means)
- *k*-nearest-neighbor
- Support vector machines

LASSO, Ridge regression:



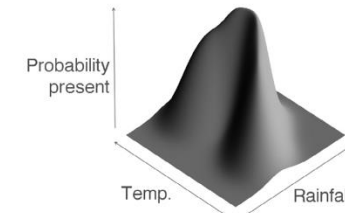
Boosted regression trees:



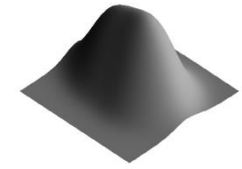
# Common ML algorithms in ecology

- Support vector machines (SVM)
- Bagging and boosting ensembles
  - Random forests (RF)
  - Boosted regression trees (BRTs)
- Bayesian-based methods
  - Gaussian processes (GP; i.e., Gaussian random fields; GRF)
  - Bayesian additive regression trees (BART)

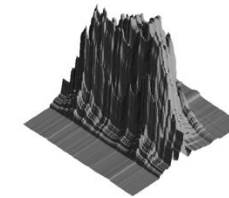
Golding & Purse 2016



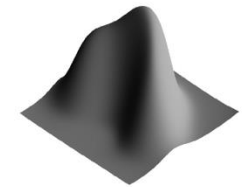
True surface



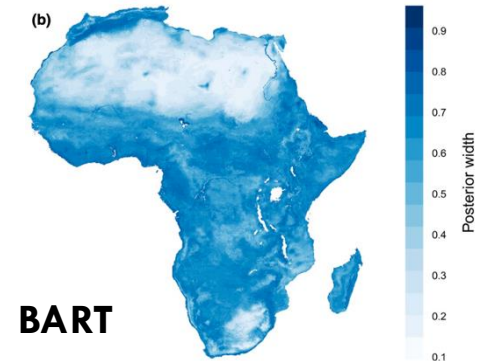
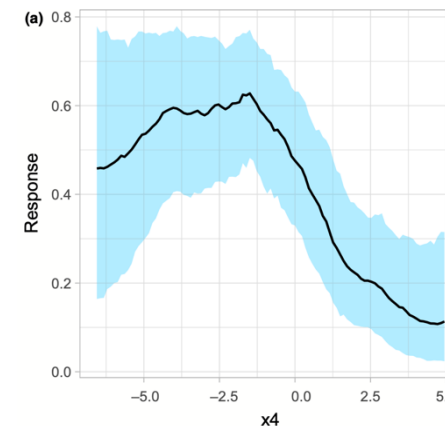
GAM model



BRT model



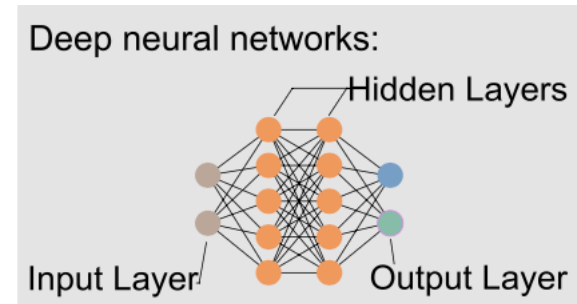
GP model



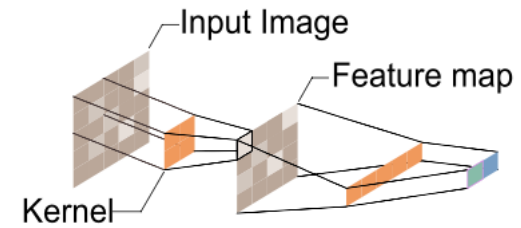
Carlson 2020

# Deep Learning (DL) algorithms

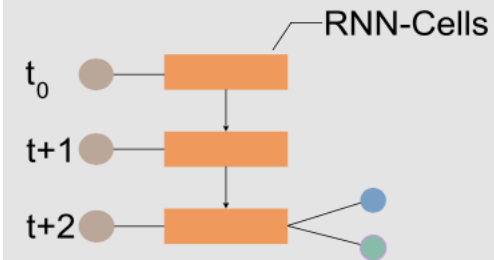
- Advancement of ML that uses multilayer neural network architecture
  - Often can include millions of parameters
  - Can address tasks often not possible with classical statistics



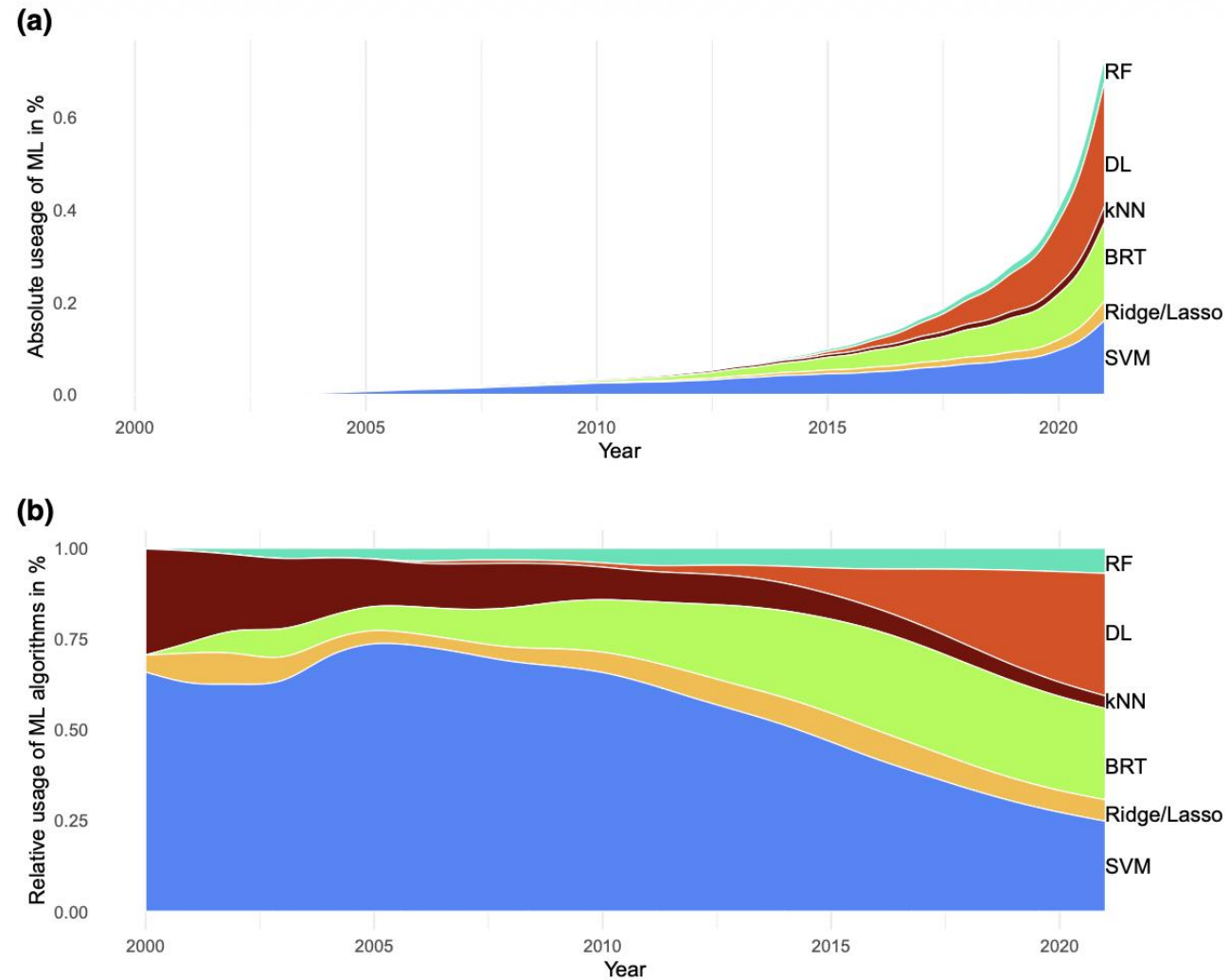
Convolutional neural networks:



Recurrent neural networks:

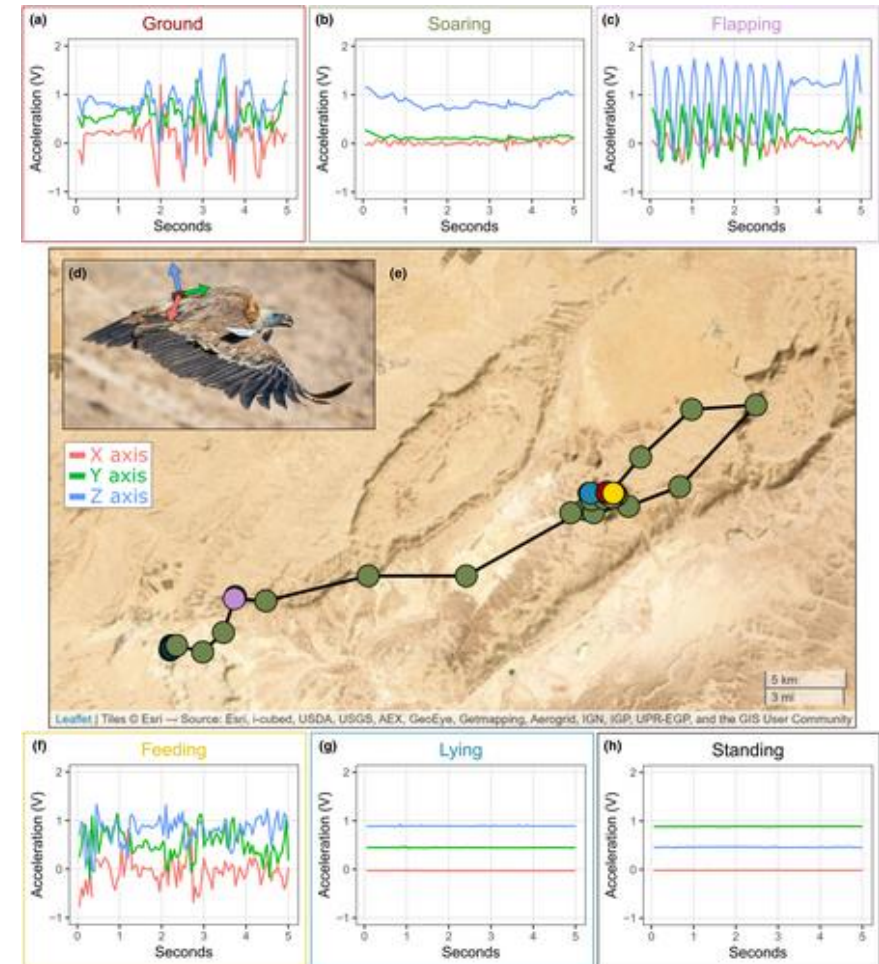


# Current trends in ecology and evolution



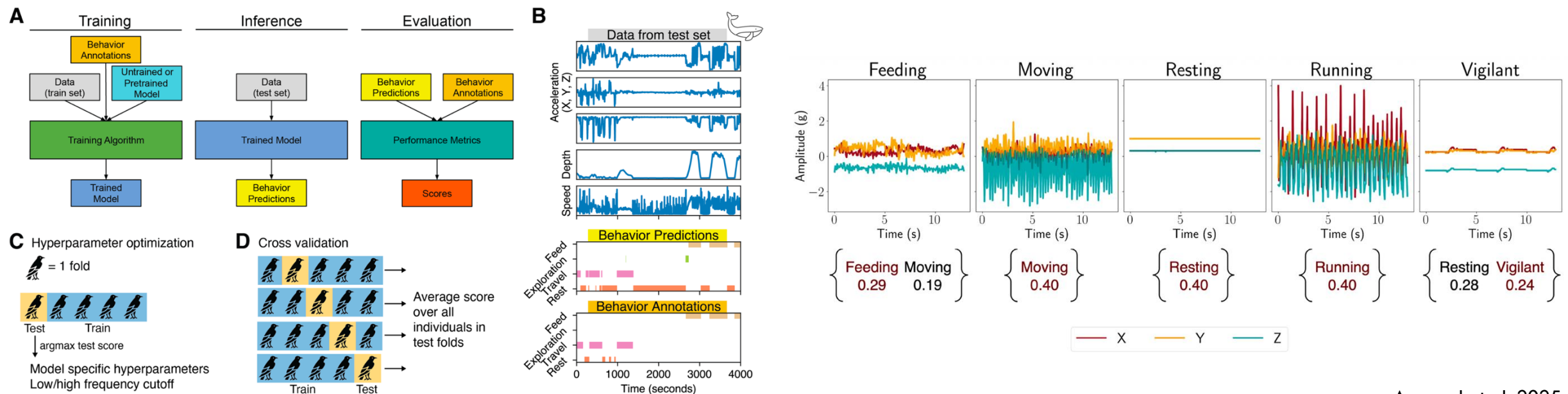
# Ecological applications of ML

- Behavior classification
- Movement-informed habitat selection
- Path reconstruction
- Species distribution modeling
- Real-time informatics
- Species and individual identification
- GeoAI



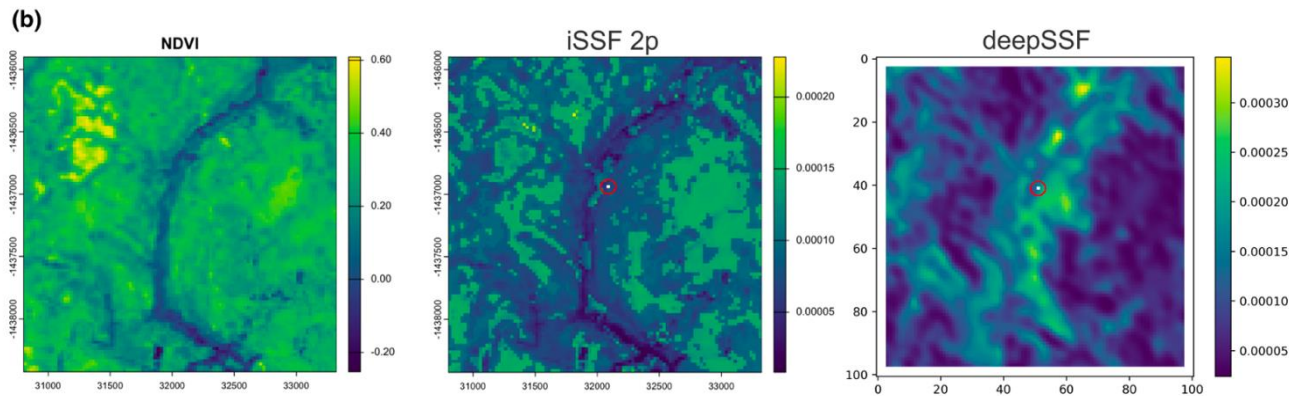
# Behavior classification

- Typically performed using dataset sampled at high resolution  $> 1$  Hz
  - Most common is triaxial accelerometer
- Common now to have labeled data for neural network training, testing, and validating

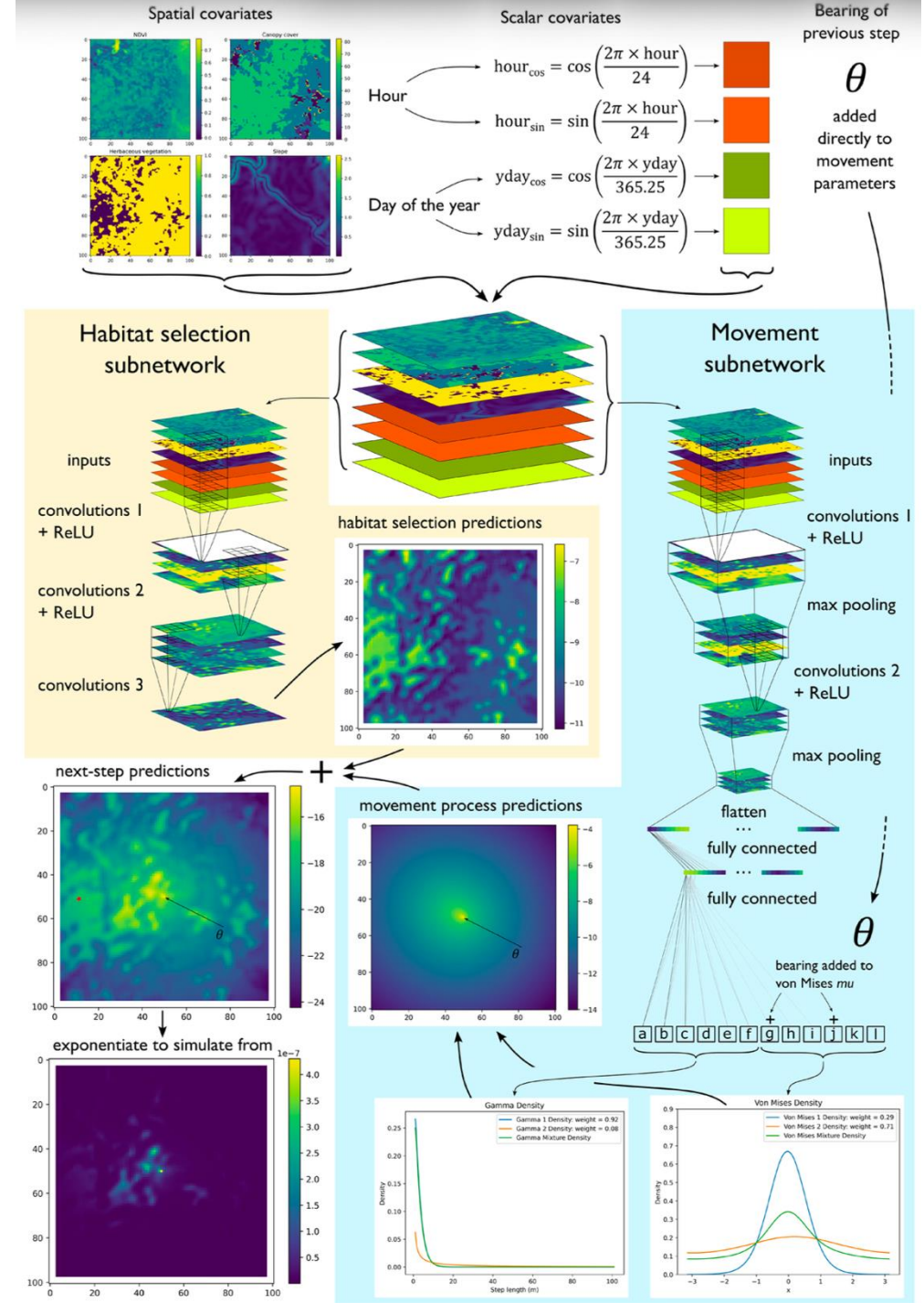


# DL-based SSF

- Handles complex species movement and habitat selection interactions without specification of the functional forms
- Can allow integration of accelerometer and other sensor data

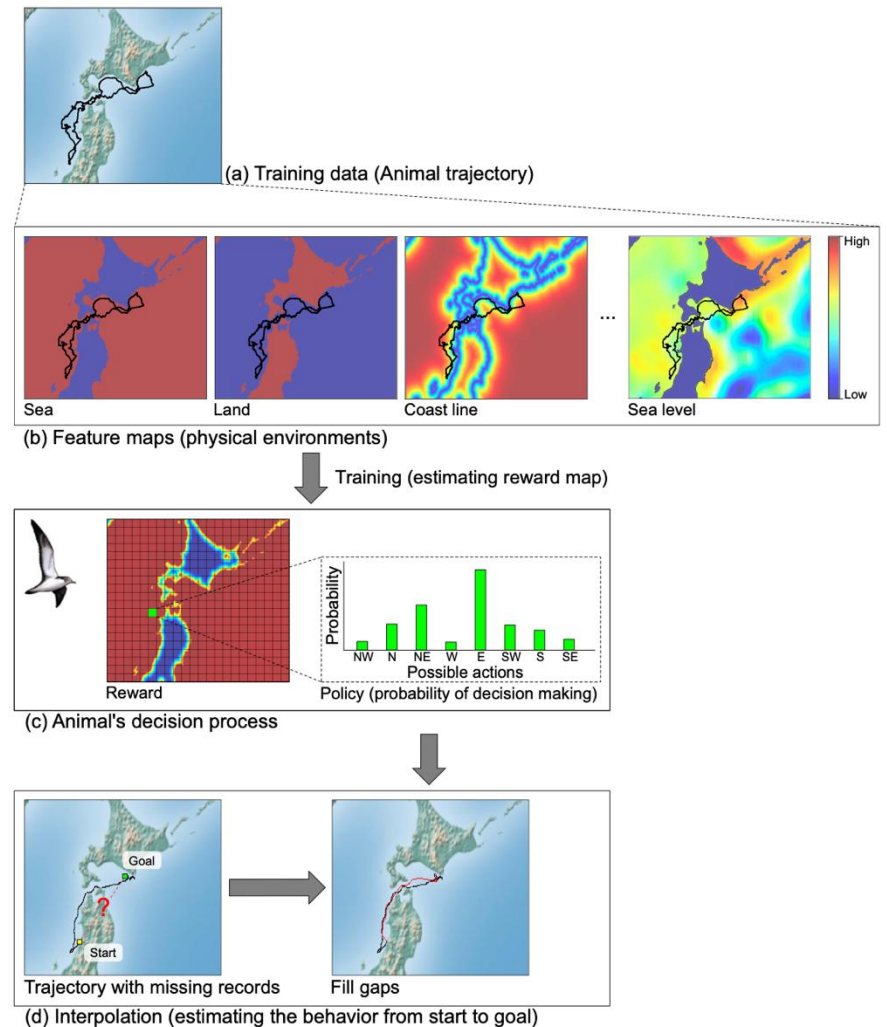


Forrest et al. 2025



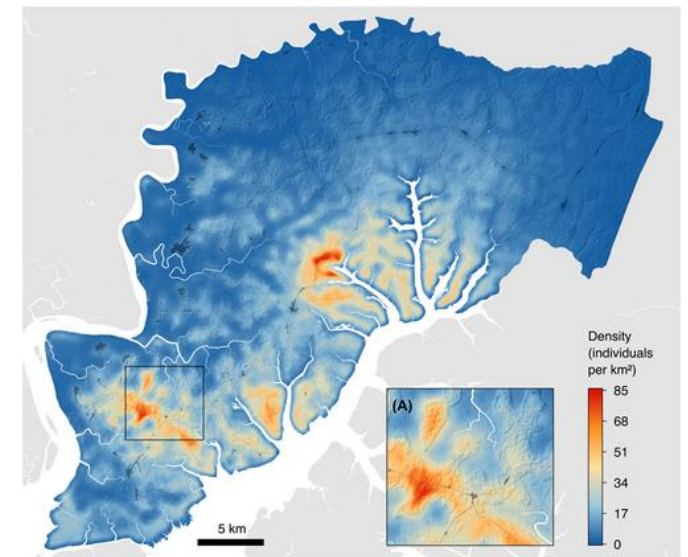
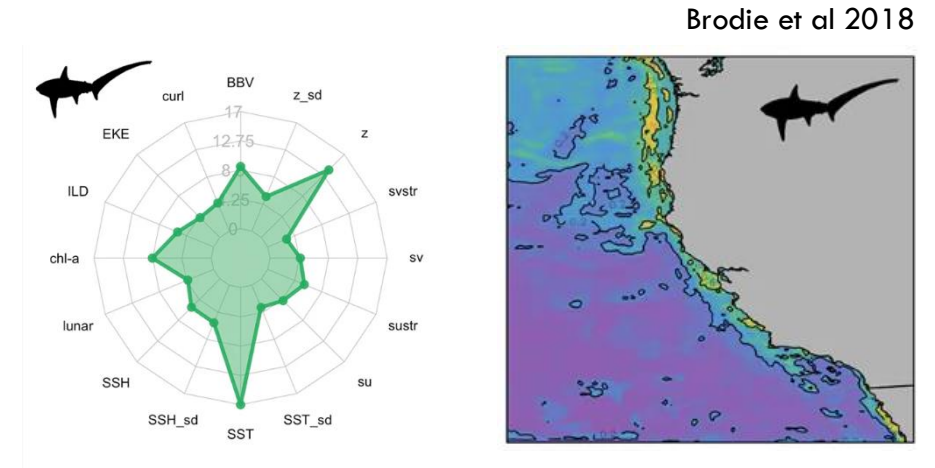
# Interpolation of telemetry gaps

- One promising example is inverse reinforcement learning (IRL) per Hirakawa et al (2018)
- Algorithm learns a reward function from animal trajectories to find the environmental features preferred by the animal
- Compared to SSMs, can better fill large time gaps and doesn't assume any functional forms
- Landscape preferences/restrictions learned from the data



# Species distribution modeling

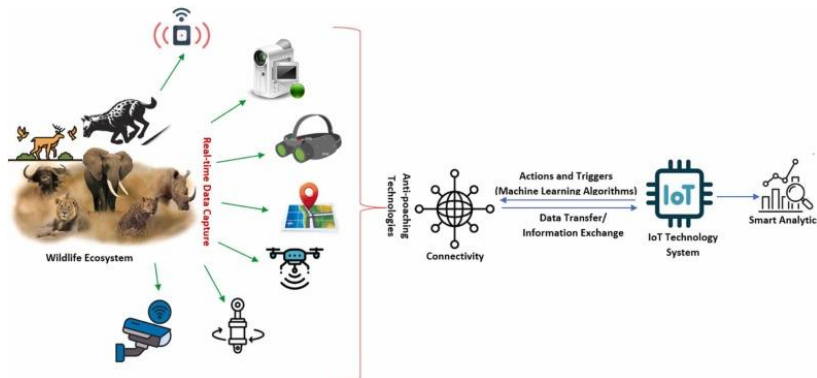
- Have been using ML algorithms for a while now
  - Primarily in marine species
  - Most common is BRT
  - But BART becoming more common
  - More studies also applying GRFs to account for spatial random effects and/or autocorrelation



Houldcroft et al 2025

# Real-time informatics

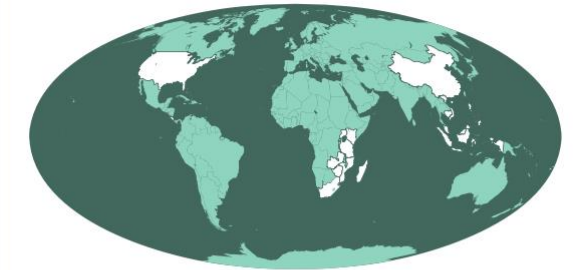
- High-throughput, multi-modal monitoring
  - Passive acoustic monitoring
  - Remote sensing
  - Geofences



Adebisi & Ndjuluwa 2026

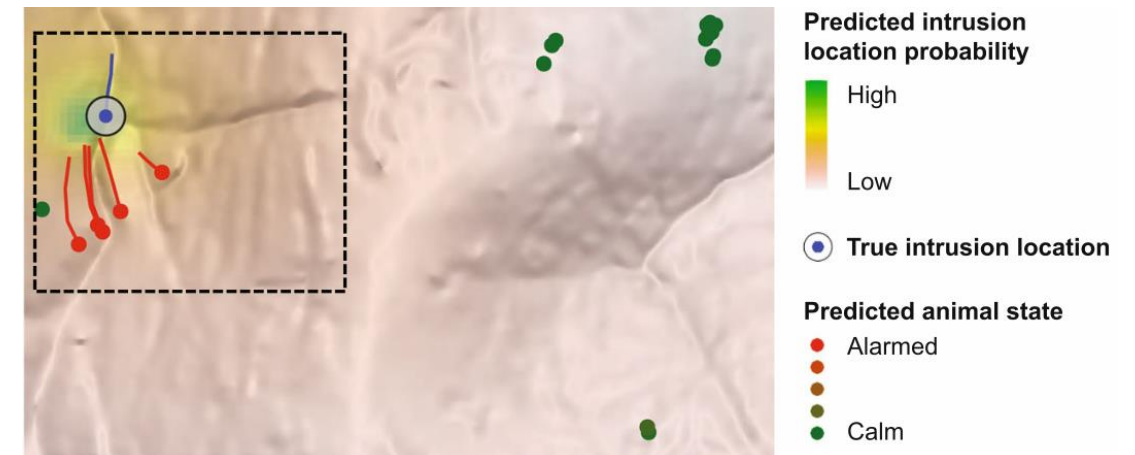


## Wildlife Protection Solutions



WILDLIFE CRIME PREVENTION  
18 nations | 800 cameras | 900K images

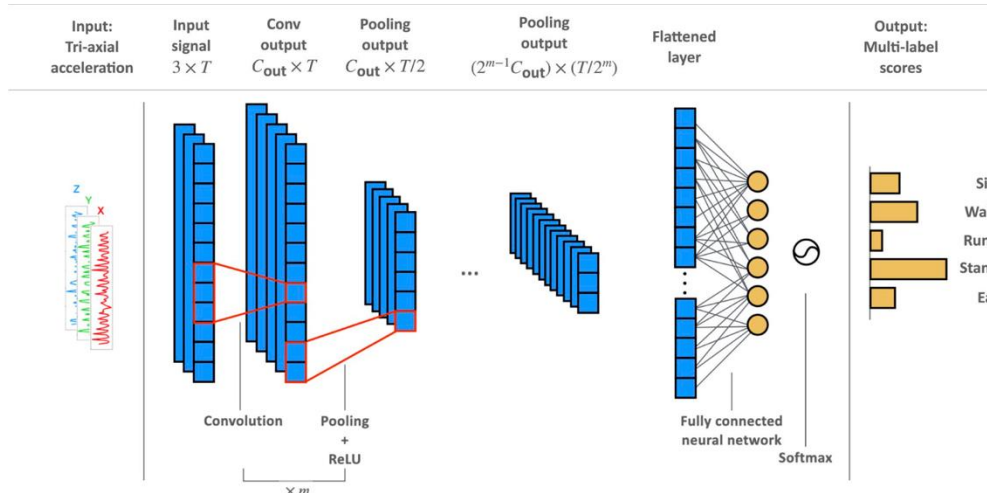
Real-time alerts  
Detects one real wildlife threat per week on average



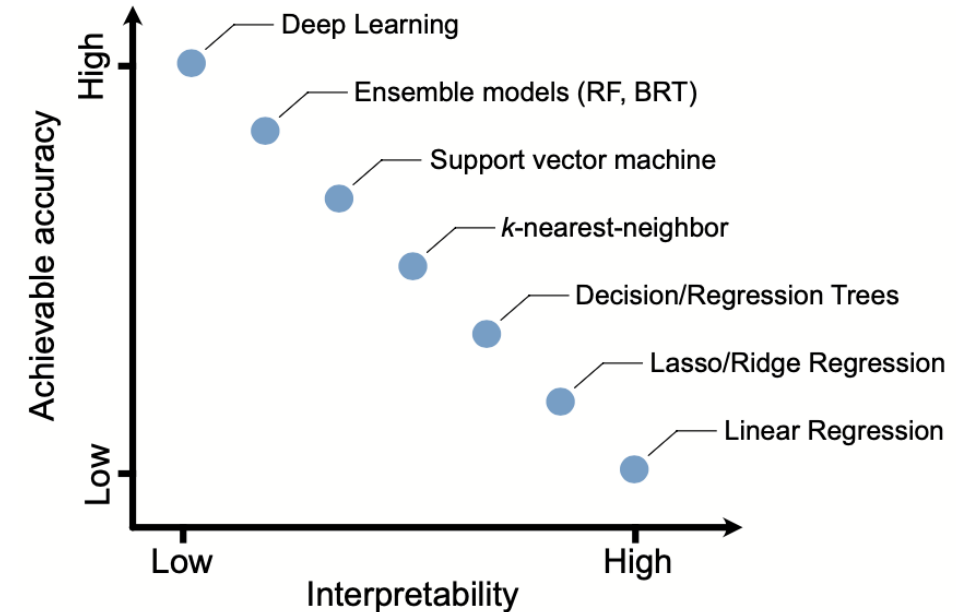
Tuia et al 2022

# The problem of ML as a “black box”

- Model complexity makes it difficult to understand what they’ve learned
- This means that ML may be better suited to prediction, but not inference



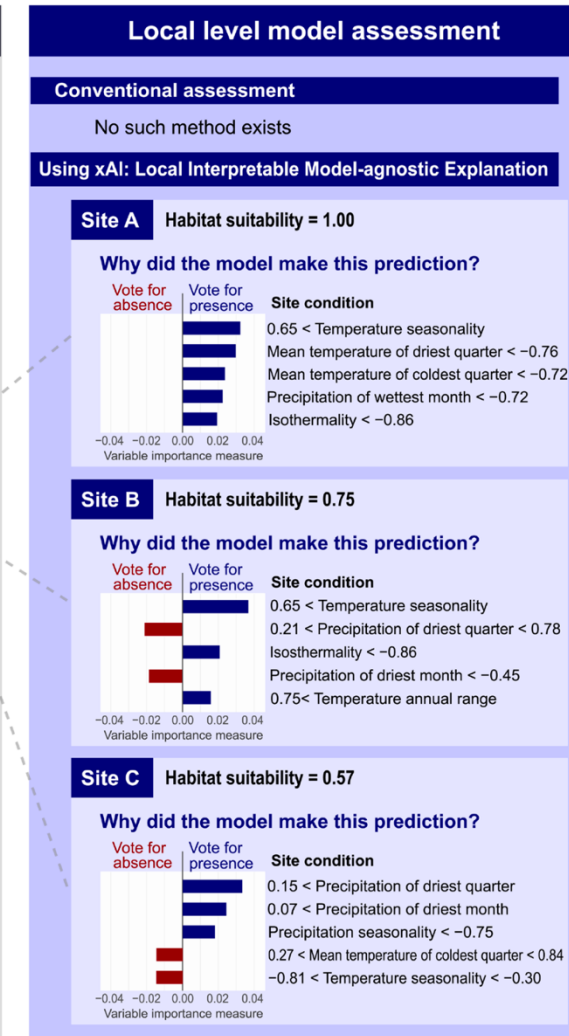
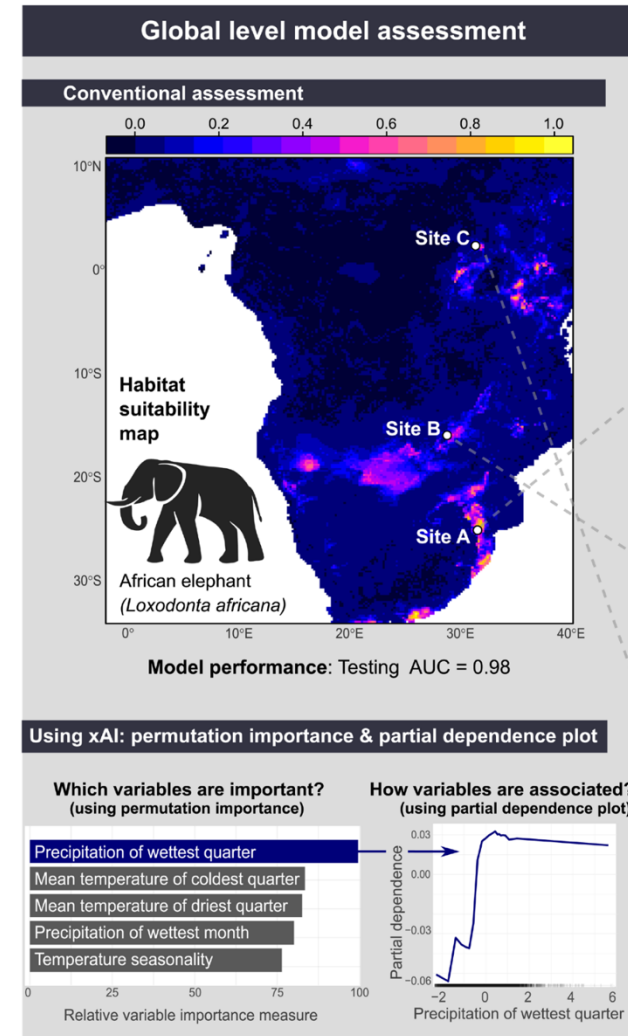
Agarwal et al. 2025



Pichler & Hartig 2023

# Explainable AI (xAI)

- Tools for providing more transparent predictions from ML/DL models
- Typically applied as *post hoc* methods
- ‘Global’ tools interpret what model learned from full dataset
  - Permutations of variable importance, partial dependence plots
- ‘Local’ tools interpret how each prediction is made
  - SHAP, LIME



# Emerging trends in ecology

- Leveraging large community science datasets
- 3D data collection and analyses to understand behavior, SHAs, and species interactions
- Scaling up of data collection across space and species
- On-board behavioral state classification for biologging devices
- Predicting and mitigating human-wildlife conflict
- Digital twins for ecosystems

